

Acknowledgements

I cannot possibly summarize my gratitude to everyone whose help has brought me closer to a PhD. My time in graduate school has been as much a social experience as an intellectual one. I feel fortunate to have grown profoundly as a person over the past five years through many experiences and interactions likely unique to this period of my life. If I am not as effusive here as I ought to be, it is only because I hope the best is yet to come in the relationships I acknowledge here.

Eric Jacobsen has shaped my experience of graduate school and my view of science in ways I probably do not yet fully appreciate. When I started learning organic chemistry, I did not expect to have the opportunity to work closely with this towering figure, much less that he would offer me unreserved/undeserved support to pursue directions of my choosing, to the extent that going to work every day was joyful and exciting. Eric's acuity, clarity of thought, and faithfulness to scientific inquiry are qualities of legend; the thrill I experienced in my first year watching him discuss science has never lessened. If anything, my admiration for his humility has only grown. From Eric, I have learned more than science—I feel fortunate to have a role model in a mentor who, above all, does everything to bring out the best in our scientist–human selves, whether by being aggrievingly patient in explaining something for n th time, or by being compassionate about circumstances outside of work. A particular highlight of my time in graduate school has been writing together with Eric, from whom I have acquired a disdain for split infinitives and an appreciation for effective communication. I only hope that I will one day feel worthy of being a small part of the Jacobsen research program. Thank you, Eric, for never once asking in my early years how my projects were going, and for giving me so much freedom and opportunities to succeed. If the one thing you will remember about me is my “elegant gluttony [*sic*],” I hope you can rest easy knowing that the calories are going to my brain (mostly, I think).

Richard Liu has been inordinately generous with his time and social energy ever since I was a green second-year. I have learned so much from attending his group meetings, if only from watching his unparalleled intuition and split-second analytical capabilities at work. I am continually amazed by the breadth and depth of his functional knowledge, and thus by the fact that he deigns at all to discuss science with me, let alone to collaborate on some of the most exciting work of my PhD. I feel fortunate to have a scientific role model in Richard and I look forward to continuing to learn from him.

Ted Betley was a source of sage advice all through graduate school. I learned to use PowerPoint animations effectively; I learned that chemical engineers not only exist but do important work, a lesson I intend to experience firsthand when I move to MIT.

It took me until embarrassingly late in graduate school to realize how much of our group's (and my own) survival hinges on **Nicole Minotti**. I feel so fortunate that Nicole bravely interfaces with the full bureaucratic machinery of this institution so that we can focus on research, meet deadlines, and remain funded. Long after I leave, I will find myself unconsciously reaching for the chocolate jar on the group room table after lunch, a testament to how much cozier Nicole has made this lab.

Professor **Scott Miller** is the reason I am an organic chemist. He welcomed me as freshman in college and helped me discover my love for catalysis through a series of profoundly exciting projects, including my first mechanistic study and some that, in hindsight, were well above my pay grade. (I'm thinking of the dynamic kinetic resolution of stereodynamic quinone–hydroquinones; I'm still not sure I can get it to work!) Throughout COVID lockdown, Prof. Miller gave me—an undergraduate—an uninterrupted hour of his time virtually every week, sitting patiently through my amateur “deep-dives” into computational chemistry. While minimally informative for him, these reviews stoked my deep interest in this area that I continue to explore. He has remained extraordinarily generous with his advice and boundless optimism through uncertain times. I feel lucky to move through my career with an incredible supporter in him.

At Yale, Professors **Seth Herzon** and **Jonathan Ellman** taught me most of what I know about physical organic chemistry. They welcomed my visits to their offices with loads of questions, which led me to discover the joys of extracurricular thinking and independent inquiry.

I am indebted to the mentors who taught me practical organic chemistry. In chronological order: **Ding Long Hoon** and **Xin Zhou** from Prof. **Yixin Lu**'s lab, **Aaron Featherston**, **Gavin Coombs**, and **Chris Yuk-Cheung Chan** from the Miller Group. Working at the hood does wonders for my happiness and wellness, and it is only due to their instruction that I can find a flow state in lab today. I first learned computational chemistry (*ab initio*, none of the DFT funny business) in Prof. **Peter Schreiner**'s lab, where **Henrik Quanz**, **Markus Schauermann**, and **Dennis Gerbig** were particularly patient and generous with their deep technical knowledge, which provided a solid foundation from which I could learn and apply plenty of other computational techniques. I hope to run a CCSDT(Q) calculation again someday.

In graduate school, I have been fortunate to find in my mentors both scientific collaborators and lasting friends. **Gabriel Lovinger** was, I have told anyone who would listen, the best thing to happen to me in my first year of graduate school. A staunch supporter and an extraordinary mentor in the philosophy of science and life, his unmatched generosity with his time meant that I always had a source of positive, affirming energy from an experienced guide as I first navigated the rocky terrain of graduate research. He remains the most inspiring resilient person I have ever known, and I cannot imagine the path I would have taken had he not fortuitously agreed to take me under his wing. **Eugene Kwan** taught me plenty of lessons about prioritizing good science, working with precision, and taking time to ruminate. I have benefited greatly from Eugene's kindness, his openness to sharing thoughts and ideas, and his reliably immediate replies on everything from KIEs to papers on biocatalysis. Both have since become people I can call in a pinch, which is more than I could ever ask for.

My greatest joy in graduate school was working among talented, driven, and genuinely kind colleagues in the Jacobsen Lab. Every one of my questions was taken seriously, and I benefited from constructive feedback at every turn. Each of my coworkers from the Jacobsen Lab has made a meaningful impact on my development as a scientist; I regret that I cannot be exhaustive in the thanks that follow.

Peyton Beyer was a pleasant surprise. I never expected to cultivate my most cherished friendship in graduate school with a garish equestrian Midwesterner from Iowa sporting an oily golden mane, who also happens to be a convicted felon in the state of Minnesota.¹ At work, he is an insightful—and crucially, honest—coworker from whom I have learned a great deal of chemistry. Outside of work, Beyton introduced me to many colorful and questionable pursuits—Goodwill- and sidewalk-hunting, David Lynch et al., psychedelic rock, craft beer, L. Ron Hubbard (the last one is a joke). I look back fondly on our spontaneous frisbee sessions on warm summer afternoons, Tuesday treks to the theatre, and the mini road trip to see Philip’s house. He has supported me through my worst while asking nothing in return, for which I am unreservedly grateful. He is the only person whose tardiness I will tolerate, which was itself a lesson. Benton brought out my inner child, manifesting in nonsensical pranks and antics that terrorized our labmates and that were definitely antipodal to productivity. My son: It fills me with pride to see you find the love and calling you deserve, and I am so eager to watch you continue growing into manhood.

Diego Diaz has shown me through example that grace, generosity, and unfettered scientific drive are entirely synergistic. He is always game for long walks through New York, during which we discuss science and life, and partake in extraordinary food; I feel lucky that despite living in different states we manage to see each other regularly, sometimes even on a whim. **Michael Nielsen**, my “buttmate” in lab for two years, embodied an enviable balance between scientific productivity and a rich personal life, a revelation for my younger self that continues to inform how I set my own priorities. Michael enrolled me in a

¹ *State v. Beyer*, No. A23-0263, 2024 (Minn. Ct. App. Jan. 29, 2024)

comprehensive, single-student introductory survey course in death metal music that culminated in the New England Metal and Hardcore Festival, which ranks among the definitive highlights of my life (*vide supra*).

Sepand Nistanaki and **Antonio LaPorte**, my current officemates, have graciously tolerated every scientific and unscientific musing of a late-stage graduate student while injecting reliable doses of chaotic fun into my every workday, to say nothing of their own scientific excellence toward which I aspire. I am grateful to **Sepand** for being first-responder to many of my results and realizations over the past two years, and to **Antonio** for keeping the ML/computational-chemistry side of my brain deeply engaged.

My cohort, **Timur Adrianov**, **Agata Bykovtseva**, and **Jonathan Wong** are inspiring friends who pushed me to become the best scientist I could be. I am fond of many nights when I would glance around the lab and find only the four of us, still hard at work long after everyone else had left.

Dennis Kutateladze has been unreserved in his advice as a scientific forebear, and I cannot wait to see all that he accomplishes going forward. I will never understand how **Corin Wagen** achieves so much in so little time, on top of fielding my inane emails about computational chemistry. **Andre Sanchez** inspires me with his passion for science and his delivery of some of the most effective presentations I have witnessed—even at 250 words per minute—and, in the bargain, helped me rediscover my love of tennis. **Sophie Shevick** and I made regular trips to the climbing wall, during which we talked science and much else. I was the lucky recipient of her formidable spice collection when she moved. **Petra Vojackova** has been a stunning exemplar of unlimited dedication (regrettably not in an actionable way); I can only aspire to a fraction of her drive. Her impeccable baked goods inspired me to take up baking, though I harbor no illusions about approaching her standard. **Dan Strassfeld** has made an unmistakable impact on my intellectual development, having blazed the trail for many of the themes in this thesis. His detailed

feedback on my work and thoughtful answers to my cold-emailed questions throughout graduate school adds to my admiration for him, especially since we have essentially never met.

I have learned a tremendous amount from the students I have worked with closely (in reverse chronology): **L Thompson**, **Yichen Jiang**, **Maximilian Frank**, **John Rezk**, and **Frank Lee**. They put their faith in me to guide them and to share in their excitement and frustrations about research. Furthermore, they were gracious and forgiving towards a mentor who was learning on the job and who was always spread too thin. I am proud to see each of them thriving in their chosen paths.

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Dan DiPrimio has been a companion in chemistry since college who, with me, moved from New Haven to Boston. His discipline across all pursuits, including his thoughtfully maintained work–life balance, remains an inspiration. He has been a steadfast listener through my many travails and introduced me to ice skating, which has since become a meaningful part of my life outside the lab. **Dominic See** has been a great roommate and was a table tennis partner in our early years.

Over the past five years, I have enjoyed the distinct fortune of being sustained by a village of friends who have wanted nothing of me but my company. They grounded me in the world outside academia and supported me in taking daring risks both in and out of work.

Jau, **Rushy**, and **Vic** remain my dearest friends, nine years after we began a weekly dinner tradition in college that somehow endures to this day across three time zones. We have adventured together, watched each other grow, and supported each other through meandering paths in life. I look forward to

sharing a continent again someday (or, dare I hope, visiting New Orleans together), perhaps after we have all retired.

Johanna, Kevin, and Lance have been steadfast companions to my life in Boston. We have shared countless weekends in a row skating, hosting food parties, and playing tennis. Johanna and Kevin have stoically tolerated my weekly onslaught of baked goods since I began my baking routine and continue to support my myriad endeavors to organize food-centric gatherings. I feel fortunate to call on them in times of need, and I look forward to at least another two years in Boston with the crew.

Zher Yin, my only Penangite friend in Boston, listened patiently to my every gripe and accompanied me on numerous expeditions in search of good food. I aspire to his wisdom, which he assures me comes only with his advanced age.

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Angela and **Tiger**, two-thirds of the “Catalytic Trio,” were integral to my undergraduate research experience in the Miller Group. Together with **Gavin**, my first collaborator/mentor, who drove me to movies and Chick-fil-A every Thursday one summer, they showed me that research could be so much fun that I couldn't imagine doing anything else. **Tori**, whom I first met in chemistry lab, enriched my life

through wide-ranging adventures across Connecticut with **Amanda**, and continues to do so years later as the reigning mayor of Boston (air quality). I am also lucky to maintain friendships with **Jared**, **Jerry**, and **Yiming** despite their busy schedules making money, saving patients, and studying diatomic molecules while singing, respectively. **Wittman** and **Dawn** are non-chemists who not only tolerated me talking about my work but even gave me big-picture feedback that was important for the mathematics in Chapter 3. **Gillian** showed me the wonders of Nature and pushed me to buy a road bike, now an indispensable part of my life.

I feel lucky that several high school friends reconvened with me at Harvard after we dispersed for college. It has been a genuine pleasure to have the familiar faces of **Zhihan**, **Jin Ming**, and **Shengyuan** nearby and reminisce over dinners about how far we have come from Singapore. I would be remiss not to mention **Suen Rong** and **Boon Xin**, my first mentor figures in high school, who continue to take a warm and genuine interest in my career and life today.

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My family is the single greatest reason I have achieved anything in life. My parents showered me with unconditional love. My mother was my first teacher, from whom I learned everything including basic arithmetic, decency, respect, compassion, and practical common sense. I was also fortunate to receive abundant love and care from all four of my grandparents growing up. **Charmaine**, my little sister, has been the brightest ray of sunshine in my life. She has always been available despite living half a world away and is the only person who can cheer me up no matter my mood. We share an incredibly quirky but

perfectly synchronized sense of humor. I miss her tremendously when we are apart but am very proud to watch her spread her wings.